

Bank Transaction Fraud Detection – Power BI Dashboard

1. Project Overview

This project utilizes a bank transaction dataset to identify patterns of fraudulent activities. It focuses on building a data-driven Power BI dashboard that empowers analysts to detect fraud effectively using interactive visuals and DAX calculations.

2. Objectives

- Detect fraudulent transactions using transaction and balance patterns.
- Visualize key fraud indicators through Power BI.
- Provide actionable insights using DAX measures and filters.

3. Dataset Summary

- Fields: step, type, amount, nameOrig, nameDest, oldbalanceOrg, newbalanceOrig, oldbalanceDest, newbalanceDest, isFraud, isFlaggedFraud
- Size: ~1 million transactions
- Source: Simulated bank transaction logs

4. Key Problems & Solutions

Problem	Solution
Which transaction types are more fraud-prone?	Bar chart with fraud % by type
How to spot suspicious transactions?	Use scatter plots for amount vs balance drops
Who are frequent fraud actors?	Top 10 origin/destination accounts by fraud count
Are false positives common?	Use DAX to compare isFraud vs isFlaggedFraud

5. Power BI Dashboard Elements

Visuals:

- Bar Chart: Fraud % by Transaction Type

- Line Chart: Fraud Cases Over Time (step)
- KPI Cards: Total Frauds, Fraud %, Avg Fraud Amount
- Table: Top Fraudulent Accounts
- Scatter Plot: Balance Discrepancy vs. Amount

Slicers:

- Transaction Type
- Time Step (step)
- Fraud Indicator (isFraud)
- Amount Range

6. Key DAX Measures

Total Frauds = CALCULATE(COUNTROWS('Dataset'), 'Dataset'[isFraud] = 1)

Fraud % = DIVIDE([Total Frauds], COUNTROWS('Dataset'))

Avg Fraud Amount = CALCULATE(AVERAGE('Dataset'[amount]), 'Dataset'[isFraud] = 1)

False Positives = CALCULATE(COUNTROWS('Dataset'), 'Dataset'[isFlaggedFraud] = 1 && 'Dataset'[isFraud] = 0)

Balance Drop = 'Dataset'[oldbalanceOrg] - 'Dataset'[newbalanceOrig]

7. Conclusion

This Power BI dashboard serves as an effective fraud monitoring tool. It reveals fraudulent patterns, accounts at risk, and potential false positives, aiding banking institutions in minimizing fraud-related losses.